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Reformulating taxes for an energy transition

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ABSTRACT

An energy transition toward clean energy sources would reduce environmental impacts. One proposal to trigger this energy transition uses economic instruments, particularly environmental taxes. This research studies the potential impact of taxes on electricity on the environment and the economy. Using a dynamic computable general equilibrium model for Spain with energy and environmental extensions, we assess their current impact on GDP growth, energy use, and a set of different pollutant emissions. Then we propose a reform that would foster an energy transition toward clean energies and assess their economic and environmental impact. We find that only taxing the production of electricity by coal, oil, and natural gas can be better for the environment and economy than taxing all forms of electricity production in a revenue-neutral context. Moreover, the production of electricity by biomass, though considered renewable, is an important source of pollutant emissions and, in these terms, should have less importance in an energy transition.

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1. Introduction

Taxes on electricity consumption and production are a common international method of energy and environmental taxation, although in most countries they were created not to reduce pollutant emissions and resource use but to increase government revenues. In Europe, this initial objective has been gradually complemented with environmental and geostrategic goals. In particular, the increasing concerns over climate change and the need to reduce dependence on foreign energy have been powerful motivations to contain energy use.

Environmental taxes are seen by economists (since Pigou, 1920) and international organizations (e.g., OECD, 2018) as a useful tool to reduce environmental impacts associated with energy consumption and production, particularly global warming and climate change. Beyond the general objective of reducing energy use, one can argue that a well-designed tax system based on energy use can help in shifting from hydrocarbons to renewable energy sources. Environmental taxes have the general policy objective to change behaviors. They reduce the targeted environmental loads by increasing the cost of the levied good or service, which generates environmental problems while the good or service is produced, distributed, or consumed. The main underlying assumption under this argument is that the increase in the cost of the levied good or service reduces the demand. Empirical evidence shows that this is true for most goods (normal goods). The theoretical grounds and early discussions can be found in Baumol (1972) and Baumol and Oates (1971, 1988).

The European Environment Agency has long been analyzing how to implement and maximize the effectiveness of environmental taxes (European Environment Agency, 1996, 2013). However, in the European Union, there are no examples of EU-wide environmental taxes because the adoption of any fiscal measure requires the unanimous agreement of the member states. In 1992, a proposal for an EU carbon tax was discussed at length (Council Directive 2003/96/EC, of 27 October 2003, restructuring the Community framework for the taxation of energy products and electricity), which led to the approval of the European Emissions Trading System (ETS), a European cap-and-trade system, to lower greenhouse gas emissions. Energy uses continue to be mostly taxed at the national or subnational level. However, energy taxation is harmonized at the EU level, which sets minimum values for the different types of fuels (Council Directive 2003/96/EC, of 27 October 2003, restructuring the Community framework for the taxation of energy products and electricity).

Among energy taxes, taxation on electricity increases the cost of electricity, reducing demand and thereby the associated environmental burdens. The concrete effects of electricity taxation depend on the configuration of the tax. Economic theory suggests that shifting taxation toward nonrenewable electricity sources would reduce environmental pressures, but it is a challenge from a macroeconomic perspective to empirically assess these effects.

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The electricity market for final consumers has been liberalized since 1997 in Spain, and both firms and households can choose their electricity suppliers. Some of them actually supply electricity generated only by renewable sources, but consumers pay taxes on this electricity the same as they do for electricity from nuclear and hydrocarbon-based sources. As consumers react to prices, it is expected that taxing the consumption of electricity coming only from nonrenewable sources, instead of all sources, helps consumers to choose low-carbon options and cause an energy transition in the mid/long term (Wissema and Dellink, 2007; Sovacool, 2013; Wang et al., 2018; Ross, 2018). Regarding the effects of energy taxes in the context of an energy transition, Sovacool (2013) explored the factors on Denmark's successful approach in reducing its dependence on foreign sources of energy to zero. He found different factors: energy efficiency, prolonged taxes on energy fuels, electricity, and carbon dioxide, and incentives for combined heat and power (CHP) and wind turbines. Zou et al. (2016) found for China that the introduction of carbon taxes shifted energy production technologies away from high carbon content fossil-fuels toward low carbon content fossil-based and renewable energies. Other studies, like Nguyen (2007) and Mondal et al. (2010) evaluated future energy technology options integrating carbon and energy tax policies, finding a range of possible co-benefits for domestic energy supply. Regarding Spain, some studies have analyzed the impacts of energy or carbon taxes using different methodologies (Manresa and Sancho, 2005; Labandeira et al., 2009; Labeaga et al., 2008; Freire-González and Ho, 2018a, 2018b), but not in the long-term context of renewable energy transitions.

In this research, we use a dynamic computable general equilibrium (CGE) model with an energy and an environmental sub-model to test the economic and environmental effects of a reformulation of electricity taxes in Spain. We also discuss how a new configuration of electricity taxes could promote an energy transition to renewable sources and the mid/long-term potential economic and environmental effects of different potential scenarios. Section 2 details the taxes on electricity in Spain, Section 3 provides the methodological details of the study; Section 4 explains the scenarios and includes the results obtained, and Section 5 highlights the main conclusions of the research.

2. Taxes on electricity in Spain

Spain imposes two different taxes on electricity, both at the national level: excise duties on electricity and a tax on the production value of electricity. Both can be considered environmental taxes because they are charged on electricity and increase its price, thus reducing demand. Both taxes are earmarked and go to the general budget.

2.1. Excise duties on electricity

The excise duties on electricity are an indirect tax on the consumption and on the supply of electricity for consumption, as well on as the consumption by the producers of electricity generated by them. It is a tax levied throughout the Spanish territory, and it is regulated in *Ley 38/1992*, *de 28 de diciembre*, *de Impuestos Especiales* (Law 38/1992, of December 28, on Excise Duties), along with other excise duties such as the taxes on alcohol, tobacco, hydrocarbons, coal, and vehicle registration.

In this section, we describe the main characteristics of this tax as it is currently defined.

Article 93 establishes that the consumption by generators or groups of generators of total power not >100 kilowatts (kW) they produce will not be subject to the tax. Article 94 defines the exemptions of the tax.

Article 97 defines the tax base as the same as that of the value added tax (VAT), excluding the electricity tax itself. Article 98 states that the net tax base will be the result of applying, if applicable, to the taxable base a reduction of 85% on the amount of electric power that is dedicated to any of the following uses:

a) Chemical reduction and electrolytic processes
b) Mineralogical processes
c) Metallurgical processes
d) Industrial activities whose electricity consumed represents >50% of the cost of a product
e) Agricultural irrigation
f) Industrial activities whose purchases or consumption of electricity represent at least 5% of the value of production

Article 99 defines the tax rates to be applied:

1. Without prejudice to what is established in Section 2 of this article, the tax rate is 5.11269632%.[1]
2. The quotas resulting from the application of the tax rate set in paragraph 1 may not be less than the following amounts:
   a) 0.5 euros per megawatt-hour (MWh), when the electricity supplied is destined for industrial uses; and
   b) 1 euro per megawatt-hour (MWh), when the electricity supplied is destined for other uses.
3. For the purposes of the provisions of Section 2, the following electricity supplies for consumption are considered for industrial use:
   a) Those supplied in high voltage; and
   b) Those supplied in low voltage for agricultural irrigation.

Article 101 states that taxpayers must fully transfer the amount of the tax on the person who carries out the taxed transaction, provided that this transfer is in accordance with the provisions of the law, whatever the existing stipulations between them.

2.2. Tax on the production value of electricity

Spain also imposes a tax on the production value of electricity, regulated in *Ley 15/2012*, *de 27 de diciembre*, *de medidas fiscales para la sostenibilidad energética* (Law 15/2012, of December 27, on Fiscal Measures for Energy Sustainability). This is a direct tax that levies the production activities and the supply to the power system. This tax is also levied throughout the Spanish territory.

Article 6 defines the tax base as the total amount received by the taxpayer (i.e., the owner of the power stations) for production and sale to the grid for each facility during the tax period. In the calculation of the total amount, remunerations foreseen in all the economic regimes deriving from the provisions of Law 54/1997, of November 27, of the Electricity Sector, will be considered in the corresponding tax period, as well as those provided for the specific economic regime for the case of production activities and incorporation into the power system.

Article 8 sets the tax rate at 7%.

3. Methodology

We have developed a highly detailed social accounting matrix (SAM) for the Spanish economy for 2007, which contains information on the excise duties on electricity. To assess the economic and environmental impacts, we use a dynamic CGE model with an energy and environmental extension calibrated for the Spanish economy for 2007. We briefly describe below the SAM developed and the main aspects and assumptions of the model.

[1] This is the tax rate in 2018. In 2007, it was 4.864%.

3.1. Data

3.1.1. Economic data

The SAM is the main source of economic data we use for our analysis. It is a square matrix that provides information on the different flows of payments among the agents and other elements of the described economic system: commodities, industries, capital, labor, land, households, firms, government, taxes and tariffs, investments, savings, and the exterior sector.

We employed supply and use tables to differentiate between commodities and industries. The data come from the National Statistics Institute of Spain (INE) and Exiobase (Tukker et al., 2013; Wood et al., 2015). Exiobase is a dataset that contains a multiregional input-output framework for 2007 in its second version and environmental information for 163 industries and 200 commodities for all the EU countries and other regions. To maintain the detail that Exiobase provides and homogeneity with other macroeconomic information from official sources (especially INE), we built new supply and use tables by breaking down electricity and waste treatment industries with the detailed information available in Exiobase tables for Spain.

After adjustments, our SAM has 101 industries and 101 commodities, with high detail in the areas of electricity generation technologies and waste treatment technologies. Regarding electricity production industries, we have information on electricity produced by coal, gas, nuclear, hydro, wind, petroleum, biomass and waste, solar, and other means, as well as electricity transmission services and distribution and trade services. Such a disaggregation is not common in other SAMs and CGE models, so they cannot focus the analysis on taxing specific technologies of electricity production.

Other information to complete the SAM comes from the INE and from the Spanish government and ministries, including a high level of detail on different taxes: VAT, sales taxes, other taxes on products, tariffs, VAT on imports, subsidies, and excise duties on alcohol, tobacco, hydrocarbons, electricity, and retail hydrocarbons. This information is particularly relevant for our analysis because we focus on taxation reforms. This information and social security accounts come from the General Intervention Board of the State Administration (IGAE).

The EU KLEMS project on growth and productivity provided data on the stock of capital and depreciation rates by industry (Jäger, 2016). Information on the foreign sector comes from the INE and firms' accounts from the Bank of Spain.

3.1.2. Environmental data

The model contains two physical sub-models: an energy and an environmental module.

The energy module contains information on coal (in tons), oil (in liters), natural gas (in cubic meters), and electricity (in GWh) production, distribution, and use. Economic data are provided by the SAM, and data on their consumption have been obtained from IEA (2016).

Data for the environmental module were obtained from Exiobase and include the combustion and non-combustion emissions of 31 different pollutants, including three greenhouse gases: CO_2 (carbon dioxide), CH_4 (methane), and N_2O (nitrous oxide). The other pollutants related to human health include PM_{10} (particulate <10 μm), SO_2 (sulfur dioxide), NO_x (nitrogen oxide), $PM_{2.5}$ (particulate <2.5 μm), NH_3 (ammonia), CO (carbon monoxide), benzo(*a*)pyrene, benzo(*b*)fluoranthene, benzo(*k*)fluoranthene, indeno(1,2,3 *cd*)pyrene, PAH (polycyclic aromatic hydrocarbons), PCBs (polychlorinated biphenyl), PCDD_F (dioxins), HCB (hexachlorobenzene), NMVOC (non-methane volatile organic compounds), TSP (trimethylsilylpropanoic acid), As (arsenic), Cd (cadmium), Cr (chromium), Cu (copper), Hg (mercury), Ni (nickel), Pb (lead), Se (selenium), Zn (zinc), SF_6 (sulfur hexafluoride), HFC (hydrofluorocarbons), and PFC (perfluorocarbon).

3.2. The dynamic energy–environment economic model

In this section, we describe the main aspects of the dynamic energy–environment economic model developed to assess the effects of electricity taxes. More technical details of the model can be found in Freire-González and Ho (2018a, 2018b). Despite the many changes, improvements, and adaptations it has experienced, the model is originally based on the work of Dale Jorgenson and colleagues (Cao et al., 2009, 2013; Ho and Jorgenson, 2007; Jorgenson et al., 2013; Jorgenson and Wilcoxen, 1993).

Four different agents are represented, and their behavior is modeled: households, government, firms, and the rest of the world. They all receive and spend income and have other roles modeled by the equations. Households, as an economic agent, basically supply labor, buy commodities, pay taxes to the government, and generate savings from unspent income. They receive income from various sources: labor, capital and land, the government, and other countries. Firms have the objective of producing goods and services by using intermediate and primary inputs from other industries. They use their income to deliver dividends to households and taxes to governments and retain part as savings. Governments basically use different forms of taxes, subsidies, and transfers to redistribute incomes within the economic system. Besides this role, the government also purchases commodities, makes investments, and pays interest on its debt. The debt is generated from government deficits, covered by domestic and foreign borrowing. The rest of the world purchases commodities (exports), provides commodities (imports), invests, and transfers income in both directions. There are five factors of production: labor, capital, land, energy, and intermediates. The production technology of industries is specified with Cobb-Douglas functions with constant returns to scale. The capital input is rented from the total capital stock that also evolves with new investments (grows) and with depreciation (declines). Labor mobility across sectors and its supply depend on the level of unemployment. The land factor is specified only for agriculture, forestry, and fishing industries, with its supply fixed exogenously.

In its dynamic aspect, the economic model is a Solow growth-based model where economic growth is driven by savings. Savings come from different agents represented in the model: households as a share of their incomes, firms retaining part of their profits, and the foreign sector investing through foreign investment. These savings are used for either financing the government deficit or for investing in domestic capital. Economic growth depends not only on savings but also on other drivers: the rate of population growth, capital accumulation, and technical change. Technical coefficients can evolve if there is technical progress (i.e., obtaining more output with the same inputs) or biased technical change (i.e., changes in input demands not related to prices).

The CGE model includes two additional sub-models, one describing energy use and another describing the emissions of pollutants. Energy and pollutant emissions coefficients are obtained from economic and environmental information described in Section 3.1.

4. Scenarios and results

In this section, we assess the macroeconomic and environmental impacts of both types of taxes on electricity in Spain: the excise duties on electricity and the tax on the production value of electricity. Then we find potential ways to improve the global configuration of both taxes to achieve better economic and environmental results, as well as to facilitate an energy transition toward renewable energies.

4.1. Scenarios developed

We have set five different scenarios with three main purposes: (1) to assess the effects of current electricity taxes (scenario A); (2) to assess the effects of a taxation reform (Scenario B is set to show the impact of reallocating taxes such that only nonrenewable sources are taxed);

and (3) to assess the effects of the reform under different potential energy transitions (scenarios C, D, and E show different potential energy transitions triggered by the proposed reallocation of taxes).

The last three scenarios embody the assumption that sustained taxation on nonrenewable technologies would create enough incentives to reduce their share in the economic structure over the years. Because we do not have data on this industry-specific effect, we set three potential scenarios: an increase of 50%, 100%, and 150%, respectively, on the use of renewable energy sources 20 years after the reform is implemented. Thus, we can observe the trends in economic and environmental effects of different hypothetical energy transitions and compare them to scenario B, where no transition is modeled. Renewables growth in Spain from 2004 to 2014 was around 90%, a little bit higher than the European average (Gabaldón-Estevan et al., 2018). This is due to many different factors. As there is some uncertainty on what would be the evolution of renewables after a tax reform like the analyzed one, we set the three different potential scenarios of renewables increase, consistent with current trends in Spain and with the European targets established in the Renewable Energy Directive (COM(2016) 767 final/2). The European Commission approved in 2016 a revision on the Renewable Energy Directive to meet the target of at least 27% renewables in the final energy consumption in the EU by 2030. For Spain, there is a specific target set at 20% at 2020. With the proposed scenarios for Spain and the information obtained from the SAM matrix developed, we obtain these figures: i) 50% renewables increase: it means renewables in 2027 would be 24.94% of total energy use; ii) 100% renewables increase: it means renewables in 2027 would be 33.26% of total energy use; iii) 150% renewables increase: it means renewables in 2027 would be 41.56% of total energy use. Here, we detail the different scenarios:

A. *Current taxes*

This scenario assesses the economic and environmental impacts of both electricity taxes. Given that we include excise duties on electricity in our SAM, we set a 0% tax rate for all electricity sources and establish the outcome as the new base case. Then we add again the excise duties and set an additional 7% tax on the production value of electricity (because the latter was included after 2007, the year of the SAM). We run the model and compare the output of this scenario with the base case without any of the electricity taxes.

B. *Reallocation*

This scenario assesses the economic and environmental impacts of exempting renewable energy sources and ensuring that the same total revenue is obtained, but only from nonrenewable energy sources. The specific changes we make are to set a tax rate of 0% for both the excise duties on electricity and the tax on the production value of electricity produced by hydro, wind, biomass and waste, and solar resources. At the same time, we proportionally obtain their previous revenues from electricity produced by nuclear, coal, gas, petroleum, and other oil derivatives such that the total revenues from both taxes remain the same and only nonrenewable sources are taxed. Previous revenues are obtained from the SAM (for the excise duties on electricity) and from applying a 7% tax on the value of electricity production.

C. *Reallocation and renewable energy transition of 50%*

This scenario assesses the economic and environmental impacts of exempting renewable energy sources and ensuring that the same total revenue is obtained only from nonrenewable energy sources (as in scenario B), but it also includes a gradual transition of the economic structure toward more use of renewables.

We reallocate both taxes in the same way as in scenario B, but we assume a slow transition toward renewables. Specifically, we simulate the share of renewable energies by all industries increasing annually at a rate such that the total use of this type of electricity is increased by 50% in year 20 because of the tax incentives. This scenario enables us to compare the tax under an energy transition (likely to be triggered by the same tax) in relation to the case where no transition occurs because we do not have information on the specific effects of the tax on different electricity technologies used. To do this, we change the technical input-output coefficients by increasing them at 2.04% annually for all renewable electricity generation technologies while reducing proportionally the coefficients of nonrenewables such that the column sum of all technical coefficients for all sectors every year is still equal to 1.

D. *Reallocation and renewable energy transition of 100%*

This scenario is similar to scenario C, but the share of renewable energies in all industries increases annually at a rate such that the total use of renewable energies increases by 100% in year 20 because of the tax incentives. Thus, the use of renewables by all economic sectors doubles. To do this, we also change the technical input-output coefficients by increasing them at 3.52% annually for all renewable electricity generation technologies and reducing proportionally the coefficients of nonrenewables. The column sum of all technical coefficients for all sectors every year is still equal to 1.

E. *Reallocation and renewable energy transition of 150%*

This scenario is similar to scenarios C and D, but we simulate the share of renewable energies in all industries increasing annually at a rate such that the total use of renewable energies increases by 150% in year 20 in relation to the base case because of the tax incentives. In this case, we change the technical input-output coefficients by increasing them at 4.68% annually for all renewable electricity generation technologies and reducing proportionally the coefficients of nonrenewables. Also, the sum of all technical coefficients for all sectors every year is equal to 1.

4.2. Environmental and economic impacts

In this section, we show and discuss the impacts of scenarios described in the previous section. With the use of GAMS software, we ran a base case scenario using the SAM and the model described in Section 3 without the excise duties on electricity; then we ran the different scenarios and compared the macroeconomic and environmental indicators with those obtained in the base case.

4.2.1. Impacts on the environment

Fig. 1 shows the variation of different pollutant emissions in relation to the base case under the scenarios described.

Both taxes on electricity (scenario A) have an impact on all analyzed pollutants, with an annual average variation that ranges from −8.27% (for selenium) to −0.10 (for ammonia). We have an average variation of −1.80% for all pollutants. In relation to the GHG pollutants we modeled, we observe an average annual variation of −1% for CO_2, −3.81% for CH_4, −0.99% for N_2O, and −0.23 for HFC.

In scenario B, where the same total amount of electricity taxes is levied on nonrenewable energy sources, we observe that 28 out of 31 pollutants show greater annual average reductions than scenario A. In this case, we have an average variation of −1.96% for all pollutants. For greenhouse gas (GHG) pollutants, we obtain −1.03% for CO_2, −3.82% for CH_4, −1.24% for N_2O, and −0.25 for HFC, so there is an improvement for all GHG pollutants.

In scenario C, we keep the same reallocation of taxes and modeled a gradual transition of the economic structure toward more use of renewable energies such that their use increases by 50%. In this case, we observe that 24 out of 31 pollutants experience a higher annual reduction in relation to scenario A, and the average variation for all pollutants is −2.16%, slightly improved compared to scenario B. Regarding GHG pollutants, we have −1.06% for CO_2, −3.62% for CH_4, −1.59% for N_2O, and −0.26

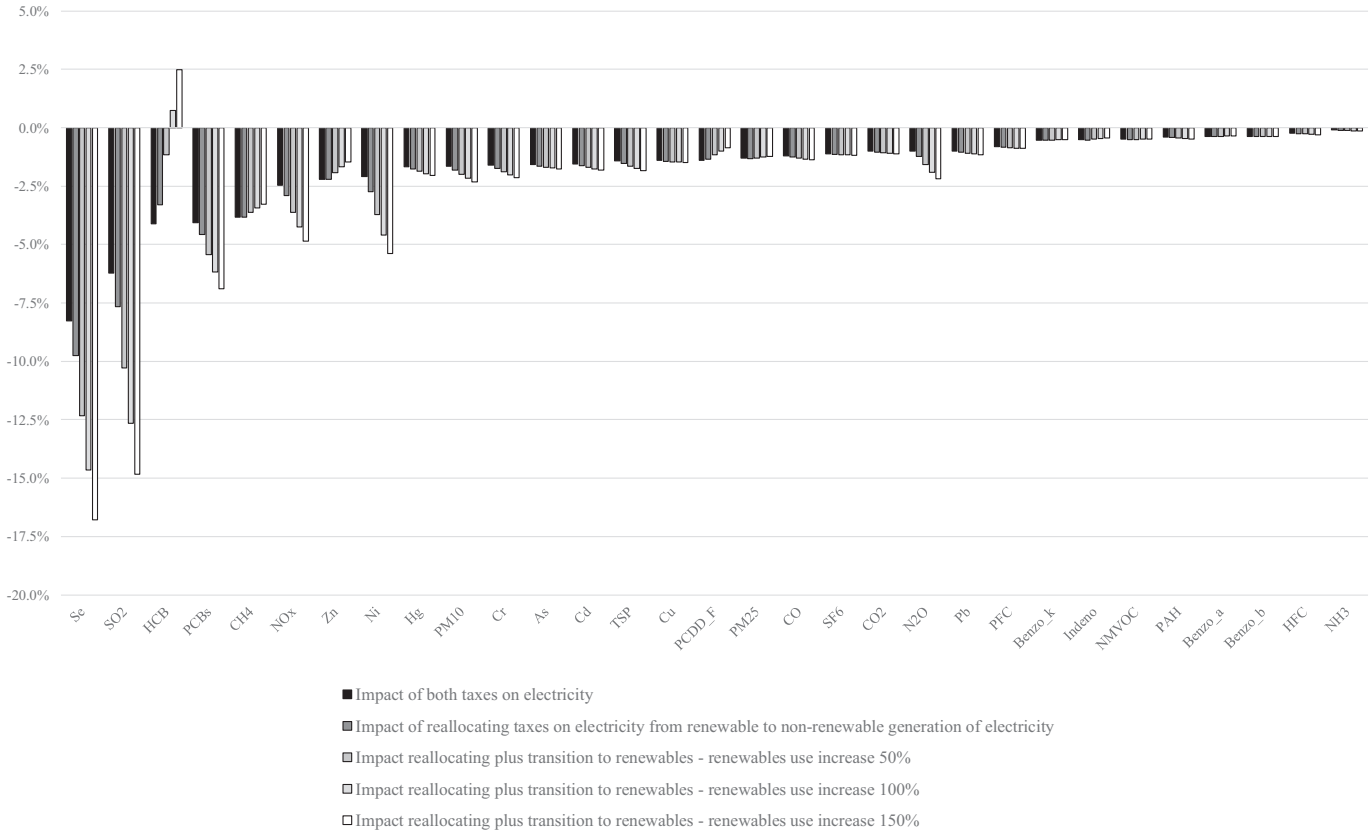

Fig. 1. Average pollutant emissions variations (for 20 years) under different scenarios in relation to the base case.

for HFC, so there is an improvement of CO_2, N_2O, and HFC in relation to scenarios A and B, but the effects on CH_4 are slightly worse.

Scenario D is similar to C, but in this case, the use of renewables after 20 years doubles. In this case, we observe that 23 out of 31 pollutants experience a higher annual reduction in relation to scenario A, and the average variation for all pollutants is −2.33%, slightly better than scenarios A and B. However, one pollutant, HCB, increases by 0.74% in relation to the base case. Regarding GHG pollutants we have −1.08% for CO_2, −3.44% for CH_4, −1.90% for N_2O, and −0.28 for HFC, so there is an improvement of CO_2, N_2O, and HFC in relation to scenarios A and B, but the effects on CH_4 are slightly worse.

Finally, scenario E is similar to C and D, but the increase in the use of renewables after 20 years is 150%. In this scenario, 22 out of 31 pollutants have a higher average annual reduction in relation to scenario A,

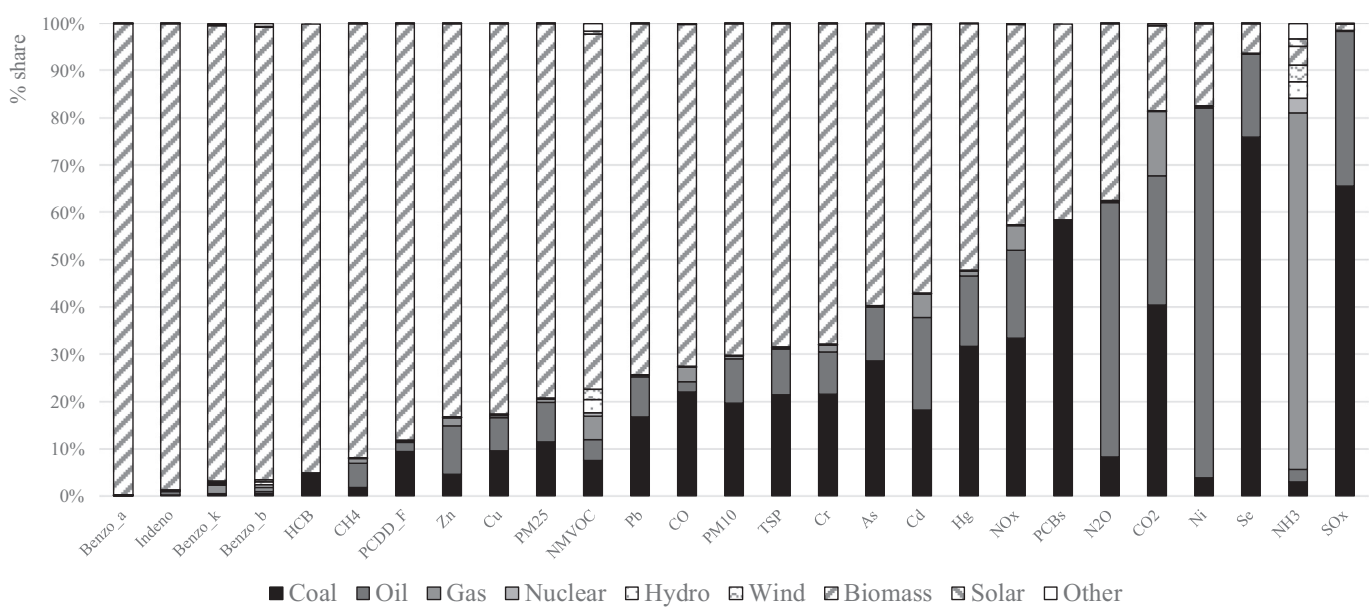

Fig. 2. Share of electricity generation technology for each direct emission coefficient.

where we analyze the current situation. The average variation for all pollutants is −2.50%, so there is a global environmental improvement. However, HCB again increases by 2.48% in relation to the base case. For pollutants directly related to climate change, GHG emissions change by −1.11% for CO_2, −3.27% for CH_4, −2.19% for N_2O, and −0.30 for HFC in relation to the base case. Thus, in this scenario, we also observe an improvement of CO_2, N_2O, and HFC, but the effects on CH_4 worsen.

It is important to highlight that although there is a general improvement, not all pollutants improve when implementing a tax reform with an energy transition. Some of them, including CH_4, show a tendency to worsen because some of them are related to the renewable energy sources. Specifically, electricity generated by biomass and waste emits higher amounts of CH_4 per unit of output in relation to the other considered technologies. Other pollutants per unit of output are also high for this technology. For this reason, the growth of electricity generated by biomass, although it is a renewable source of energy, should be limited. Although Fig. 1 shows indirect and general equilibrium effects in the variations, Fig. 2 shows the share of each technology in the direct emissions coefficients (in tons of pollutant per euros of output). We observe that electricity generation by biomass has higher direct emissions coefficients than other technologies, higher even than those based on hydrocarbons for most of the pollutants analyzed. The inclusion of biomass as a renewable source helps in understanding why not all pollutants drop when considering an energy transition. As the importance of biomass in the energy mix grows, these pollutants become more abundant. The pollutants that do not drop more as we move from scenario A to D, as shown in Fig. 1, are benzo(*a*)pyrene, indeno(1,2,3 *cd*) pyrene, benzo(*k*)fluoranthene, HCB, CH4, PCDD_F, Zn, PM25, and NMVOC; all of them rank among the first positions in Fig. 2, being mostly emitted by biomass generation of electricity rather than by other technologies.

With this undesirable effect in mind, additional measures must be implemented to control this specific technology. This can be done by implementing better ways of using taxation, like those specifically focused on externality or pollutant taxation, or by considering other normative measures.

4.2.2. Impacts on the use and production of energy

Regarding the use of energy, Fig. 3 details the average annual variation use of coal, oil, and natural gas under the different scenarios.

The current taxation on electricity (scenario A) reduces coal use by 3.35% on average and natural gas use by 0.88%. Yet it increases oil use by 0.02%. In a scenario where taxation on renewables disappears and is moved to nonrenewables (scenario B), coal and natural gas experience a higher reduction (−3.45% and −0.91%, respectively), but oil increases by 0.02% in relation to the base case. If we additionally model a transition toward an economy with 50% more renewables use (scenario C), coal and gas experience a higher reduction in relation to the base case (−3.56% and −0.95%, respectively). With a transition that doubles the use of renewables in 20 years (scenario D), coal and gas use changes by −3.66% and −0.97%, respectively, in relation to the base case, but oil use increases by 0.04%. Finally, in a transition that increases renewables by 150% (scenario E), coal and gas use decrease by 3.66% and 0.97%, respectively, in relation to the base case, but oil use increases by 0.05%.

The model also allows the analysis of energy use in physical terms. Fig. 4 shows the changes in the energy produced from domestic fossil fuels for the four scenarios, and Fig. 5 shows the changes in the energy produced from domestic renewables in exajoules.

The taxation reform we simulate in all the different scenarios reduces the production of energy from hydrocarbons more than the current taxation on electricity. If we consider the production of energy from renewables, we observe a similar pattern (i.e., the proposed reform is

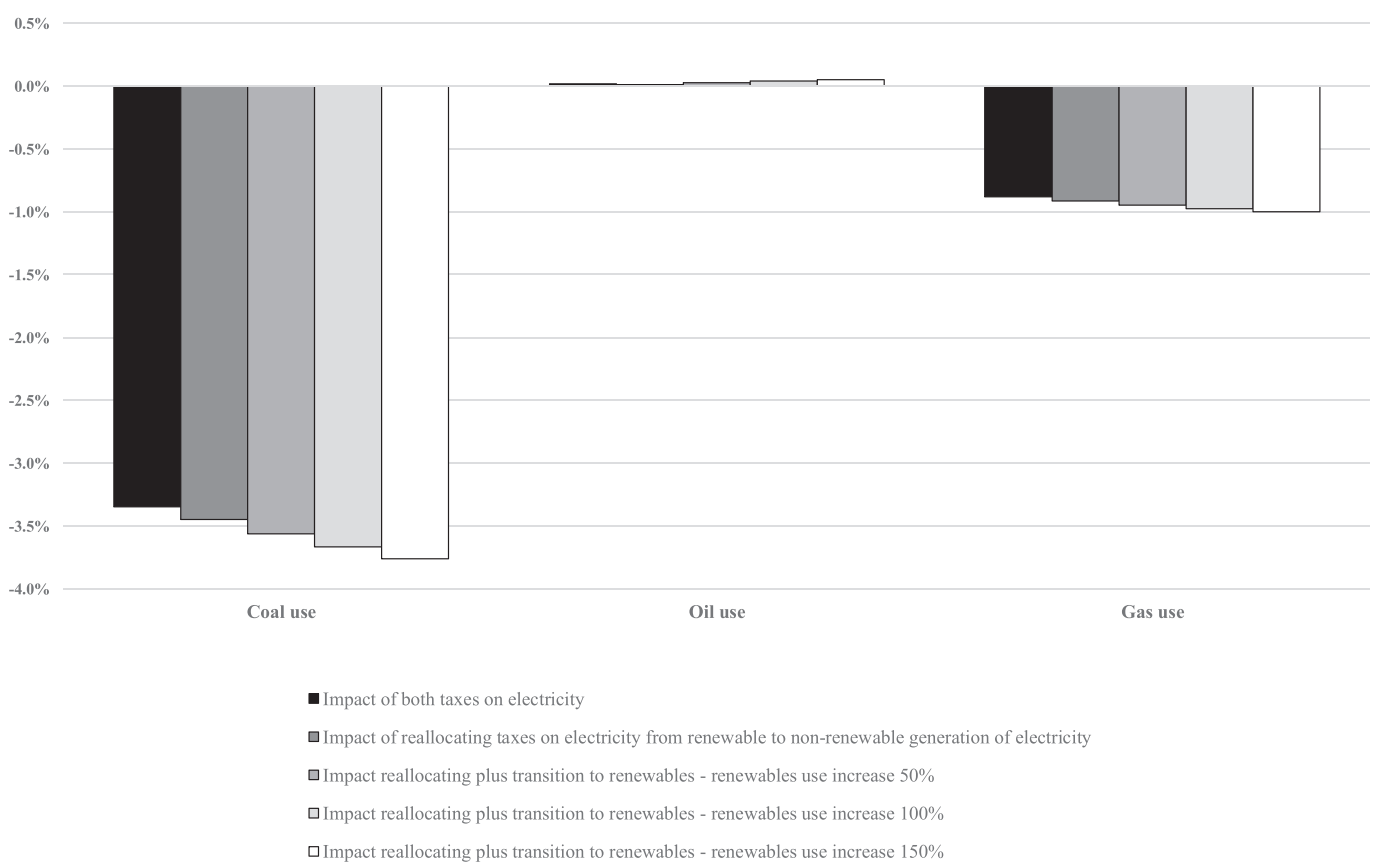

Fig. 3. Average use of hydrocarbons variation (for 20 years) under different scenarios in relation to the base case.

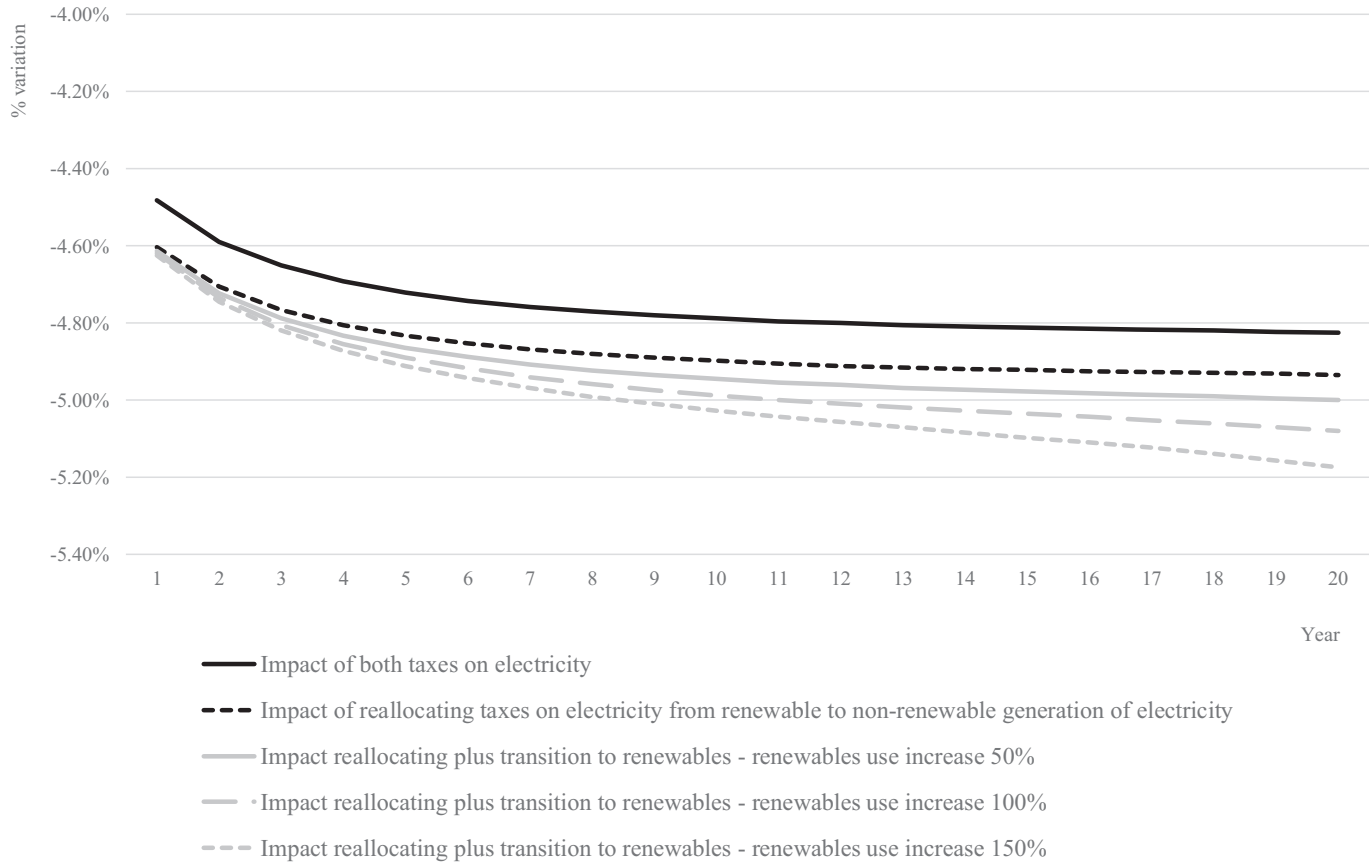

Fig. 4. Variation of energy production from domestic fossil fuels (in exajoules) under the different scenarios in relation to the base case.

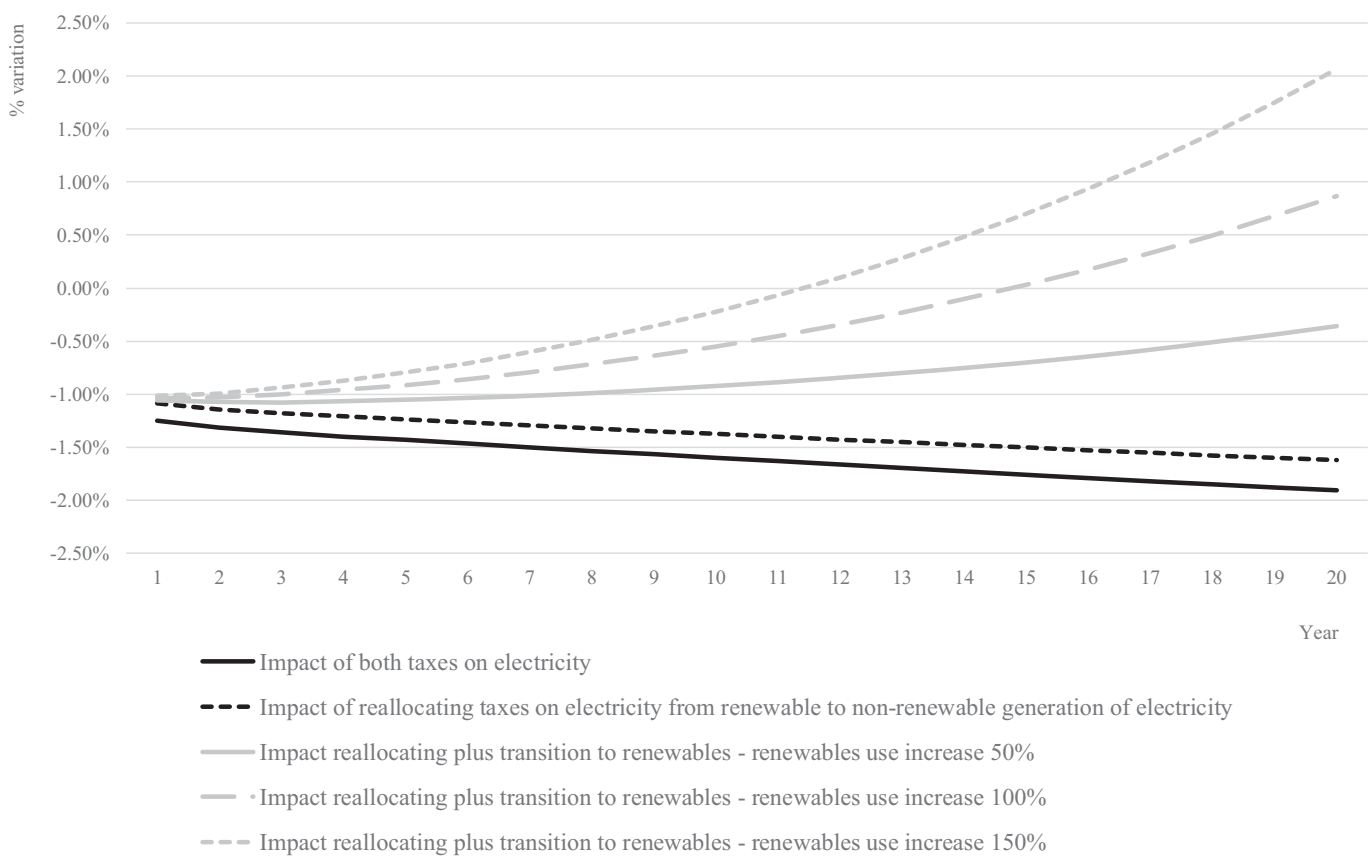

Fig. 5. Variation of energy production from domestic renewables (in exajoules) under the different scenarios in relation to the base case.

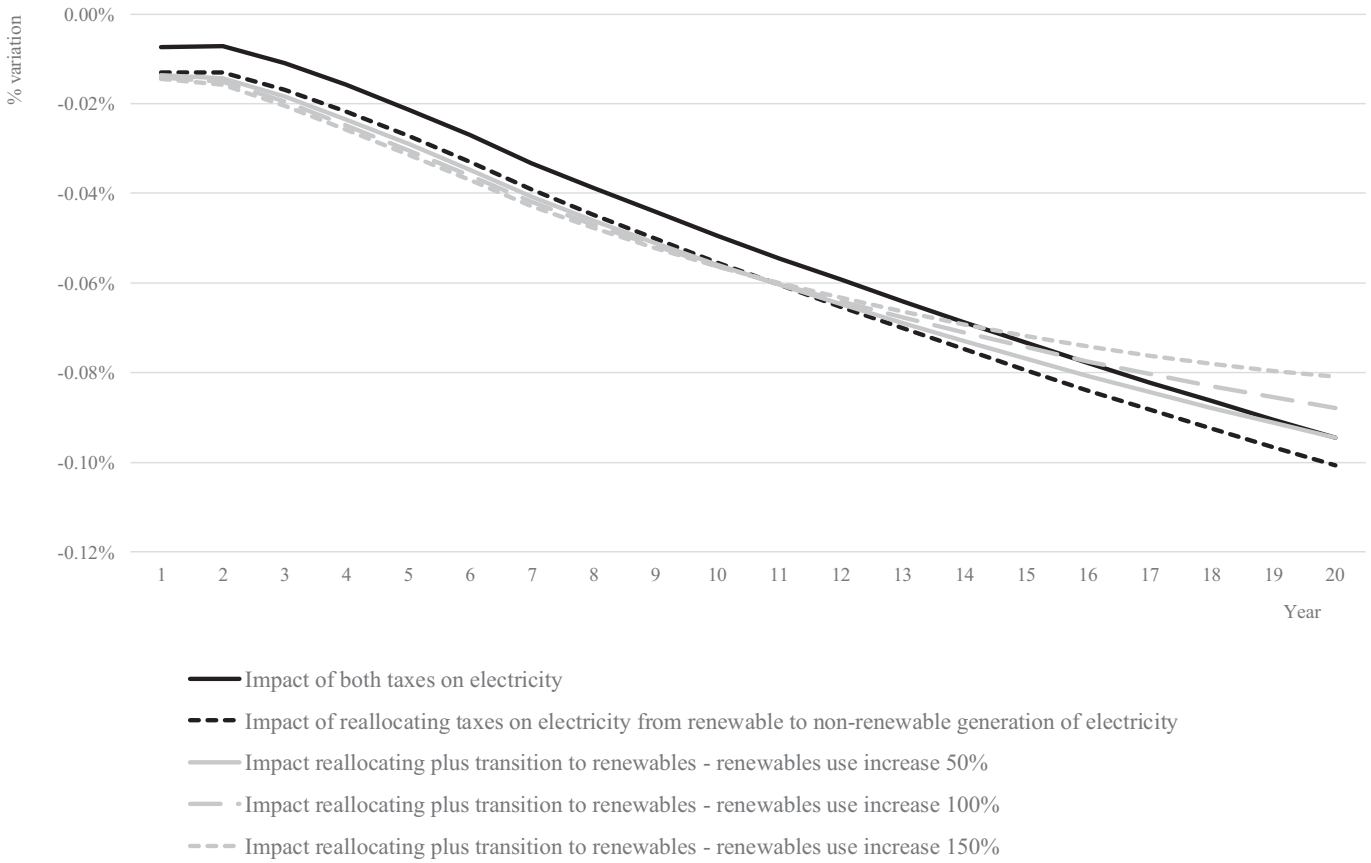

Fig. 6. GDP variation under different scenarios in relation to the base case.

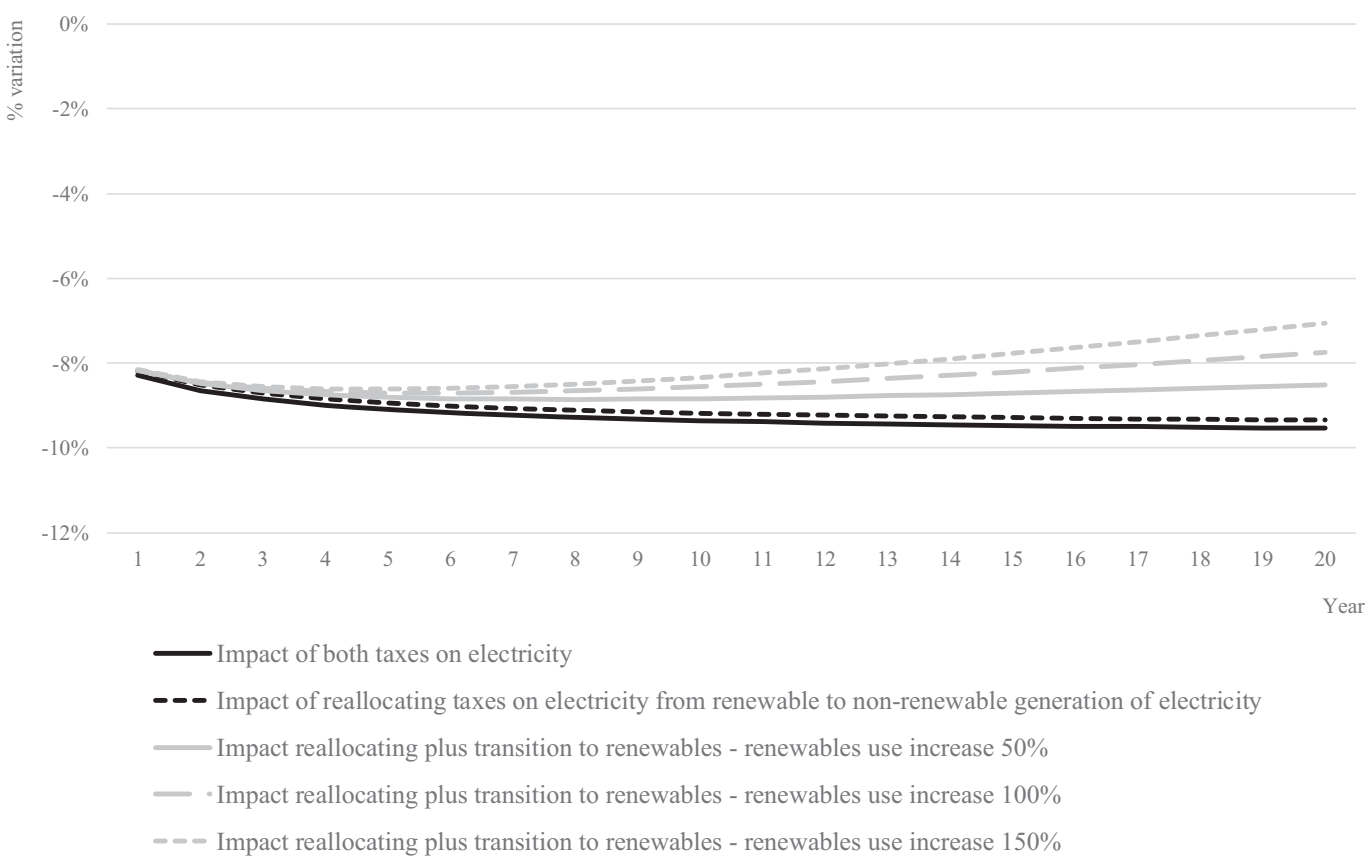

Fig. 7. Domestic production of electricity sector variation under different scenarios in relation to the base case.

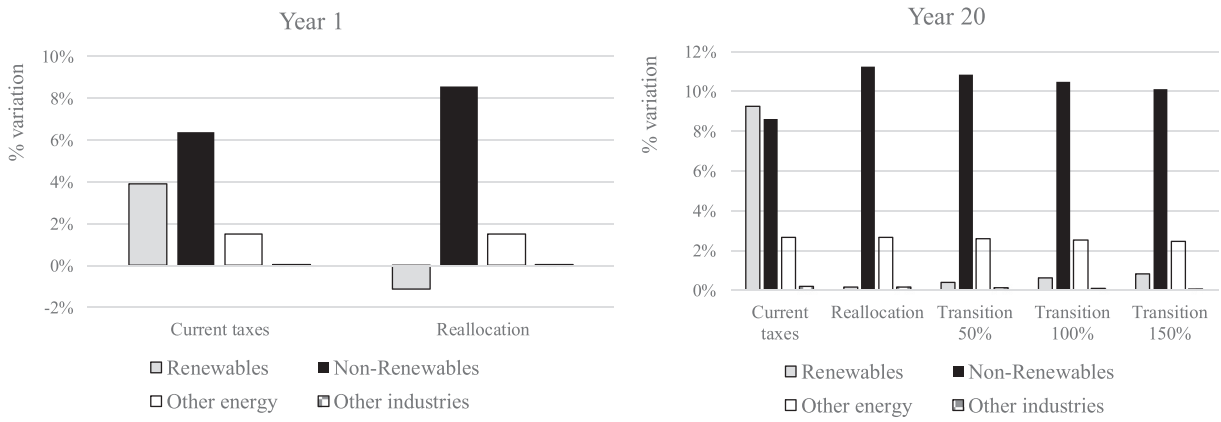

Fig. 8. Price variation of groups of industries under different taxation scenarios in relation to the base case. Year 1 and year 20.

better). Considering transitions (scenarios C, D, and E), energy production from renewables increases in the mid- and long-term in relation to the base case.

4.2.3. Impacts on the economy

Fig. 6 shows the economic effects of different scenarios in terms of variation of GDP in relation to the base case.

The economic effects under the dynamic framework show that the taxes on electricity have a negative impact on GDP and its growth; however, it is a limited impact because of the low amount these taxes represent in relation to the total GDP. We estimate with the model that current taxes suppose a GDP contraction of −0.01% for the first year and −0.09% in year 20 in relation to the base case. When we reallocate taxes (scenario B), this impact increases but is also limited. In this case, it represents a variation of −0.01% in year 1 and −0.10% in year 20. Thus, the proposed reform would slightly increase costs for the overall economic system. However, when accompanying this reform with an expected gradual transition toward renewables (scenarios C, D, and E), the economic impact is gradually reduced over time in relation to the other scenarios. These costs are low; they barely reach −0.1% of GDP at their lowest level, but we can observe tendencies of different options at higher taxation levels. These results make sense because in year 20 the taxed industries are smaller than in year 1 when assuming a transition; in other words, renewables industries are larger, have a higher share of the electricity industry, and are not taxed with the proposed reform. So, taxes have a lower impact on total GDP. However, these taxes mostly keep improving the environment, as shown in Fig. 1.

Focusing on the total production of electricity, Fig. 7 shows its variation under the different scenarios. In all scenarios, a reduction in domestic electricity production occurs, but as we move from no transition to a heavy transition, the total production of electricity decreases less because of a change in terms of trade. In heavy transition scenarios, domestic production of renewables increases in relation to non-transition scenarios by substituting imports of hydrocarbons. However, this effect is small in our model.

We can also analyze the effects of the taxes and the reforms under the different scenarios on the other industries. As we showed, we included 101 industries in our model. We have obtained the specific impact of the scenarios on all the industries. To summarize the amount of information derived from this disaggregation, we have grouped the industries into four different categories: (1) renewables containing all industries that produce electricity from renewable sources; (2) nonrenewables containing all industries that produce electricity from nonrenewable sources; (3) other energy with other industries related to energy extraction, production, or distribution; and (4) other industries containing all the other industries of the economy (not related to energy). Fig. 8 shows the price variation under the different scenarios for these industry groups, Fig. 9 illustrates production variation, and Fig. 10 shows labor demand variation.

In the first year, we observe that prices increase heavily in nonrenewable industries, especially after the proposal reform without energy transition (scenario B). Other industries are similarly affected in both

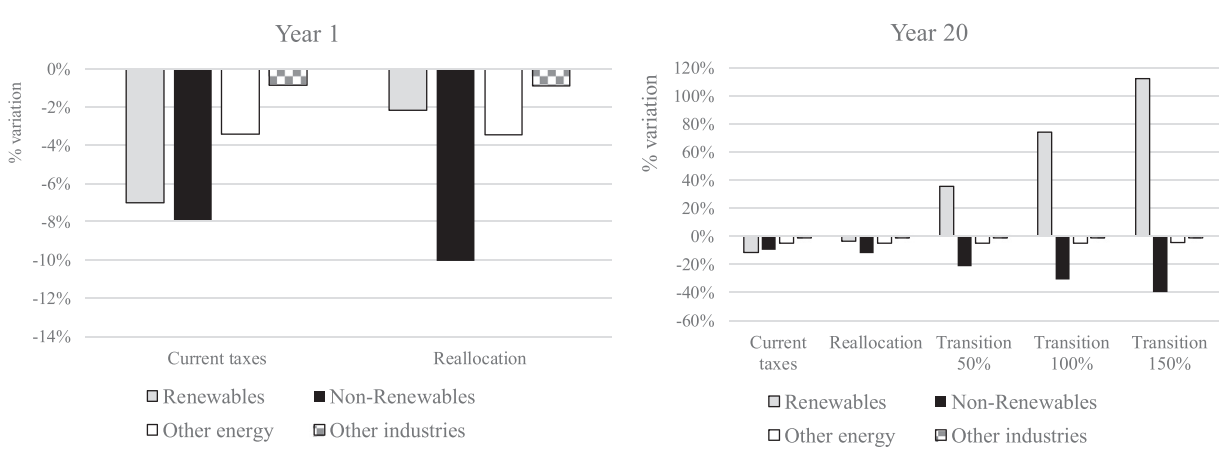

Fig. 9. Production variation of groups of industries under different taxation scenarios in relation to the base case. Year 1 and year 20.

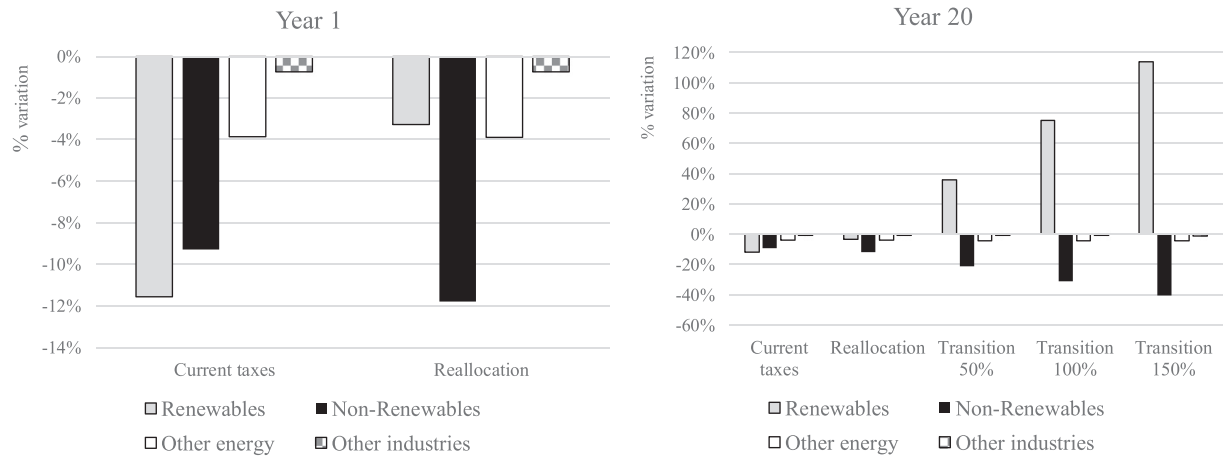

Fig. 10. Labor demand variation of groups of industries under different taxation scenarios in relation to the base case. Year 1 and year 20.

scenarios. In year 20, the current taxation on electricity (scenario A) increases the prices of renewables more than the prices of nonrenewables. However, reallocation of taxes (scenario B) has a small impact on renewable prices and a huge impact on prices of nonrenewable sources. This effect remains similar in all transition scenarios.

If we focus on industries' production, we observe that the first-year effects of scenario B (reallocation of taxes with no energy transition) tend to reduce nonrenewables more than renewables, as compared to scenario A. In year 20, we have similar effects, but if we consider an energy transition (scenario C, D, or E) renewable production increases and nonrenewable production decreases more as we move from a "soft" transition scenario to a "heavy" transition; this is the intended effect of the tax reform. Effects on labor demand are similar to those on production.

Fig. 11 shows that there is a global improvement after the reallocation of taxes, even without an energy transition (scenario B) because global production and labor demand are reduced less, and prices do not increase as much. When we assume energy transitions (scenarios C, D, and E), we observe that production and labor demand increase even in year 20 in relation to the base case.

In all these reforms, we assume that revenues generated by the government are used as public expenses in the same way the government usually uses its revenues. However, there is the possibility of using these revenues to reduce other distorting taxes such as those on labor, capital, or consumption. Empirical evidence indicates that in this way a double dividend (i.e., an improvement of the environment and economic performance) can be achieved (Freire-González, 2018). In our model, we include an additional endogenous variable (λ_t):

$$t_{jt}^{w} = \lambda_t \, t_{j0}^{w}, \tag{1}$$

where t_{jt}^{w} is the tax rate for the tax w and industry j, and the 0 subscript denotes the base case. Then we introduce an additional restriction of government spending neutrality:

$$GG_{scenario\ B}(t) = GG_{base}(t), \tag{2}$$

where GG is the quantity of aggregate government purchases. GG is endogenous in the base case and exogenous in the revenue recycling scenarios. With these additional conditions in the model, we ran an additional model with scenario B, using electricity tax revenues to reduce taxes on labor. See Fig. 12.

Fig. 12 shows that the initial negative variation of the GDP turns positive when revenues are used to reduce taxes on labor, achieving a double dividend. Thus, even in the case that scenarios C and D, which include transitions, did not emerge after the initial proposed reform (or showed

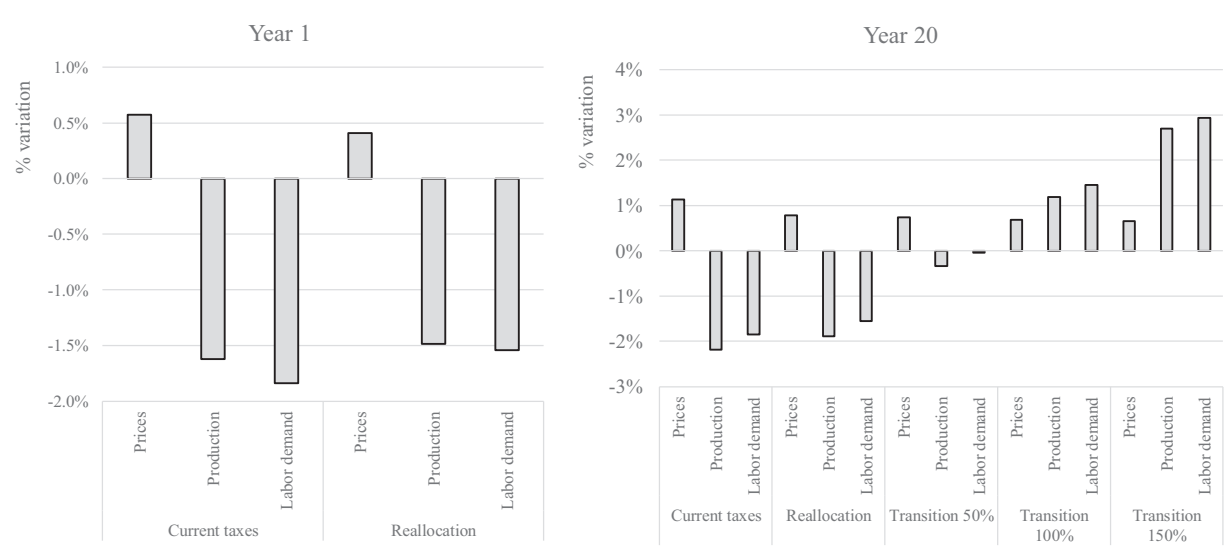

Fig. 11. Average industry variation of different economic indicators under different taxation scenarios in relation to the base case. Year 1 and year 20.

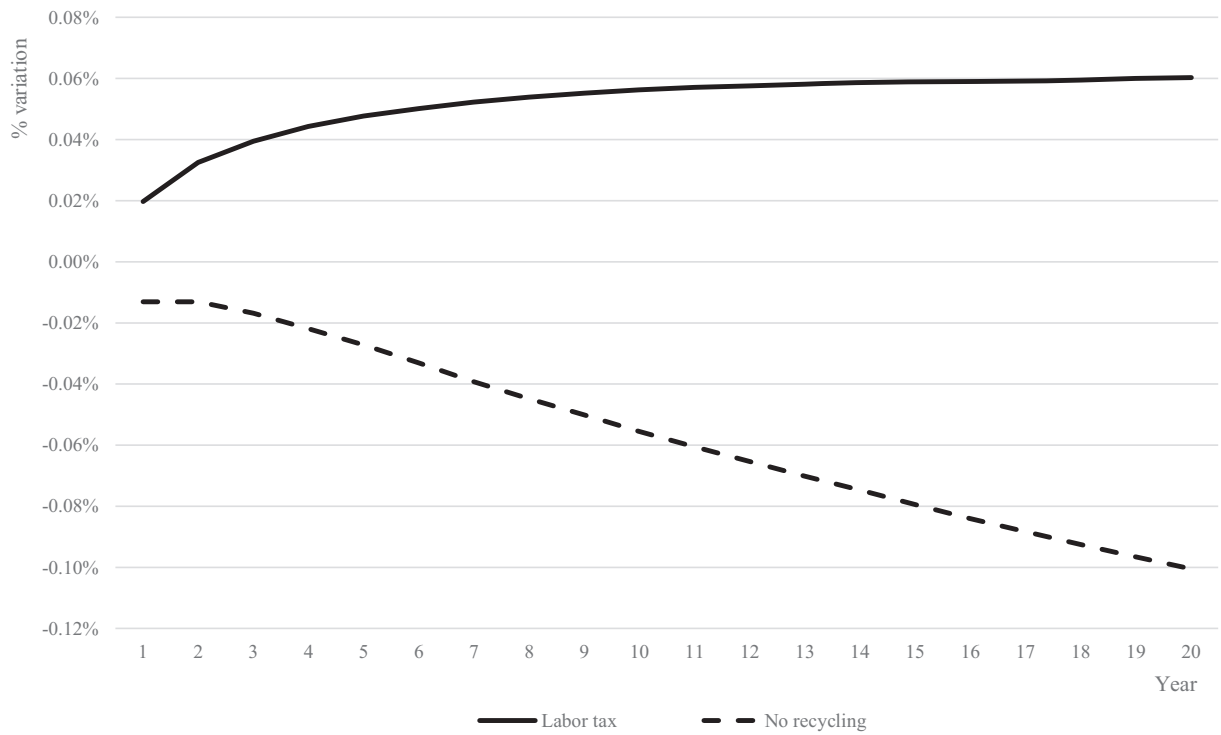

Fig. 12. GDP variation of reallocating taxes (scenario B) under two different revenue-recycling scenarios in relation to the base case.

up minimized), there is a possibility to improve the design of both taxes on electricity such that there are no costs of implementation.

We have not found other studies delivering quantitative results on the economic and environmental effects of energy taxes in a context of renewables transition. However, other collateral analyses of energy taxes suggest similar results in terms of the potential effects of taxes. Sovacool (2013) found that prolonged taxes on energy fuels was one of the factors in Denmark's successful approach in reducing its dependence on foreign sources of energy to zero. Zou et al. (2016) found the effectivity of carbon taxes for a renewables transition in China with an increase of total energy system costs under all scenarios they tested. However, results were not estimated in GDP terms.

5. Conclusions

Several policies can motivate an energy transition where clean energies become preeminent. Environmental taxation is one of the preferred policies among environmental economists. However, current environmental taxes in many countries do not have a proper design. Spanish electricity taxes do not distinguish between types of electricity production and so do not incentivize the use of clean sources instead of dirty sources. They are also conceived as ad valorem taxes, rather than as based on actual consumption, which would be more directly related with the actual impact.

We have shown that despite their current configuration, the electricity taxes currently implemented in Spain will improve the environment. However, when introducing discrimination between renewable and nonrenewable energy sources, the analyzed short-term and long-term environmental indicators improve further. We also assume an energy transition fostered by the tax reform. If we focus on the economic impacts of the reform, we observe that indicators slightly worsen in relation to the current situation, but this could be minimized depending on the chosen implementation.

We have specifically analyzed five scenarios with the use of a detailed dynamic CGE model. The first scenario shows the impacts of current taxes on electricity in Spain. In general, they are positive in environmental terms and negative in macroeconomic terms if we do not consider the benefits of reducing pollutants as an externality. In the second scenario, we propose shifting electricity taxes from renewables to nonrenewables, obtaining the same total revenues from both electricity taxes. This configuration shows an improvement in environmental indicators and worsens economic performance. However, we show in the third, fourth, and fifth scenarios that additional dynamic considerations tend to improve both the environment and the economy. The general dynamic consideration adopted in this research is that taxation on nonrenewables creates incentives for more adoption of renewables. Thus, the proposed taxation reform would help trigger a transition toward renewable energy, and this would produce better environmental and economic outcomes than the current situation in the mid- and long-term.

All observed costs in this analysis are low; they barely reach −0.1% of GDP at the lowest level. However, they help in estimating the likely tendencies of different policy options. Of course, higher levels of taxation would mean higher costs and higher pollutant reductions.

Economic results make sense because in simulations that include the fiscal reform and renewables transitions, taxed industries in year 20 are smaller than in year 1. Renewables industries grow, but because they are not taxed with the proposed reform, taxation has a lower impact on total GDP after some years of implementation. It is interesting to observe that even though total taxation drops over time (as the base tax is reduced), it mostly keeps improving the environment, partly because of the lower global use of hydrocarbons. However, not all pollutants improve when implementing a tax reform that fosters an energy transition; some of them have the opposite tendency, and it is critical to monitor the relationship of these with the transition and apply supplementary policies to control them. We have shown, in this sense, that it is necessary to apply a special treatment to electricity generation by biomass because, although it is classified as renewable, it carries a great deal of associated pollutants. We have seen that the transition should focus on the growth of the other renewable technologies (wind, solar, and hydro) to improve all pollutants in the long term. An externality tax on specific pollutants rather than charging industries or technologies

would probably provide better results in terms of environmental indicators, but their effects on stimulating a transition to renewables would be smaller.

Although the main result (taxing electricity produced from fossil fuels is better than taxing all forms of electricity production) can seem obvious, actually, in Spain and in other countries electricity taxation is homogeneous over all forms of electricity generation. Besides, we empirically demonstrate and quantify, with the use of a dynamic CGE model with environmental extensions, that taxing electricity from fossil fuels improves the environment not only by reducing the expected pollutants (GHG emissions), but also by reducing other pollutants, not as directly related to energy use. Another important finding of the research, not so intuitive, comes from the economic side: this kind of taxation can also have positive long-term effects on the economy while it boosts an energy transition.

Beyond just moving electricity taxes to nonrenewables, additional low-cost measures could be carried out to improve the economic effects of these taxes. We have shown that the use of revenues from these taxes to cut other pre-existing taxes, like taxes on labor, would result in the same environmental benefits and reduce their costs or even produce benefits in terms of GDP variation.

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